BALAKUMARAN S

Full-Stack Developer — IoT Enthusiast — AI Enthusiast

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PROFESSIONAL SUMMARY

Passionate Computer Science student with strong interests in full-stack development, IoT systems, and intelligent software solutions. Experienced in building real-world applications across web, analytics, and smart automation domains through hackathons, academic projects, and collaborative development.

Skilled in developing scalable systems using Python, Flask, Flutter, Node.js, and ESP32-based embedded platforms. Known for rapid learning, interdisciplinary collaboration, and translating practical problem statements into functional technical solutions.

EDUCATION

Sri Sivasubramaniya Nadar College of Engineering (SSN) — Chennai, India

Integrated M.Tech in Computer Science and Engineering | 2024 – Present | Semester 4 | CGPA – 9.115/10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Software Engineering, Computer Networks

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, Dart
Full-Stack & Backend: Flask, Node.js, Next.js, REST APIs, Prisma ORM
Frontend & Mobile: HTML, CSS, Flutter
Databases: PostgreSQL, MySQL, Oracle SQL, Firebase Firestore
IoT & Embedded Systems: ESP32, ESP32-S3, Arduino, Raspberry Pi, Sensors & Relay Integration
AI / Data & Tools: Streamlit, Machine Learning Basics, Git & GitHub, Linux/Ubuntu, VS Code

PROJECTS

Vendor Hiring Management Platform | *Next.js, Node.js, PostgreSQL, Prisma*

- Developed a full-stack platform for managing vendor recruitment workflows, onboarding processes, and task assignment systems.
- Designed scalable backend APIs and database schemas using Prisma ORM and PostgreSQL.
- Implemented structured role-based workflows and modern responsive interfaces.

Smart Auto Assist Home Safety System | *ESP32-S3, IoT, Embedded Systems, Sensors*

- Built an IoT-based smart home automation and energy optimization system during a 24-hour hackathon.
- Developed a lightweight analytical model for live power consumption monitoring and peak-hour alert generation.
- Designed an interactive embedded GUI using ESP32-S3-BOX for real-time visualization and user interaction.

Groundwater DWLR Monitor | *Flutter, Dart, Analytics, IoT*

- Developed a cross-platform application for real-time groundwater level monitoring and anomaly detection using DWLR datasets.
- Implemented predictive analytics and environmental data visualization modules.
- Designed dashboards for monitoring water level trends and station health metrics.

CrimeSpot — Crime Analysis & Prediction Platform | *Python, Streamlit, Machine Learning*

- Developed an AI-assisted crime analysis platform featuring hotspot visualization and ML-based forecasting.
- Integrated geospatial mapping and predictive analytics for crime trend analysis.
- Built interactive dashboards and automated insight generation modules.

Campus Resource Sharing Platform | *Flutter, Firebase, Node.js, Socket.IO*

- Built a campus-focused item lending and borrowing platform with QR-based handover verification and real-time chat.
- Implemented trust scoring, transaction workflows, and Firebase-based authentication systems.
- Integrated scalable backend communication using Socket.IO and Firestore streams.

Mess Management Platform | *Flask, Python, HTML, CSS*

- Designed a digital mess management system for handling meal scheduling, attendance tracking, and token-based management.
- Streamlined administrative workflows and improved meal coordination processes.

HACKATHONS & EXPERIENCE

Smart India Hackathon (SIH) 2025 — *Participant*

- Worked on a groundwater monitoring and predictive analytics platform using Flutter and environmental datasets.
- Contributed to system design, analytics modules, and visualization workflows.

Hack Fusion 1.0 — Jeppiaar Institute of Technology — *Inspiring Team Award*

- Participated in a 24-hour interdisciplinary hackathon focused on IoT-based smart automation systems.
- Worked on embedded analytics and ESP32-based interactive interfaces for energy optimization systems.

Cyber Hackathon — Thoothukudi — *CrimeSpot Project*

- Developed AI-assisted crime hotspot prediction and geospatial analysis platform.
- Worked on analytics dashboards and machine learning integration.

SNU Hackathon 2025 — *Campus Resource Sharing Platform*

- Built a secure campus sharing ecosystem with QR-based verification and real-time communication systems.
- Focused on collaborative application development and backend integration.

ACHIEVEMENTS & ACTIVITIES

- **Inspiring Team Award** Received in the HACK-FUSION1.0
- Participated in multiple national and inter-college hackathons across IoT, AI, and web domains.
- Active contributor to project-based collaborative development within academic peer groups.
- Strong interest in IoT systems, backend engineering, and scalable application architectures.

TECHNICAL INTERESTS

Full-Stack Development · Web Development · IoT Systems · Backend Engineering · Automation Systems · AI-Assisted Applications · Data Analytics